



Shore

Year 12

Term II Examination

May 2016

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Board-approved calculators may be used
- A BOSTES Reference Sheet is provided
- Answer Questions 1–10 on the Multiple Choice Answer Sheet provided
- In Questions 11–16, show relevant mathematical reasoning and/or calculations
- Start each of Questions 11–16 in a new writing booklet
- Write your examination number on the front cover of each booklet
- If you do not attempt a question, submit a blank booklet marked with your examination number and “N/A” on the front cover

Examination Number:

Set:

Total marks – 100

Section I Pages 3–6

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II Pages 7–14

90 marks

- Attempt Questions 11–16
- Allow about 2 hours 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the Multiple Choice Answer Sheet for Questions 1–10.

- 1 What is the value of $\frac{4.56^3 - \sqrt{78}}{\sqrt{6.8^2 \times 9.3^6}}$ correct to 2 significant figures?
(A) 0.01
(B) 0.02
(C) 0.015
(D) 0.016
- 2 Which of the following is a simplification of $4m^{-2} \div \frac{1}{2}m^{-1}$?
(A) $\frac{8}{m^3}$
(B) $\frac{8}{m}$
(C) $\frac{2}{m^3}$
(D) $\frac{2}{m}$
- 3 Which of the following represents the solution to $|2x - 3| \leq 1$?
(A) $-2 \leq x \leq 2$
(B) $x \leq -2$ or $x \geq 2$
(C) $1 \leq x \leq 2$
(D) $x \leq 1$ or $x \geq 2$

DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM

4 What are the values of a and b if $\frac{3}{2+\sqrt{5}} = a + \sqrt{b}$?

- (A) $a=3$ $b=-6$
- (B) $a=-6$ $b=3$
- (C) $a=-6$ $b=45$
- (D) $a=45$ $b=-6$

5 Let $\log_a 2 = p$ and $\log_a 3 = q$.

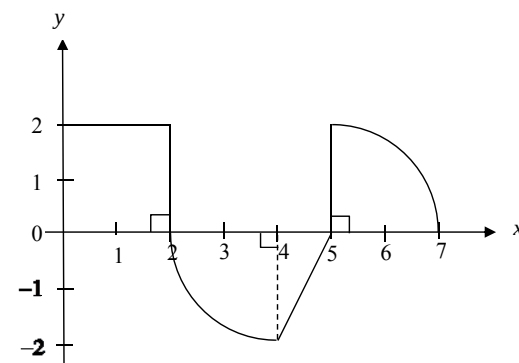
Which of the following is the expression for $\log_a 24$?

- (A) $3p+q$
- (B) p^3+q
- (C) $3pq$
- (D) p^3q

6 The roots of $2x^2 - 4x + 7 = 0$ are α and β . What is the value of $\alpha^2 + \beta^2$?

- (A) -3
- (B) 4
- (C) 11
- (D) $10\frac{1}{4}$

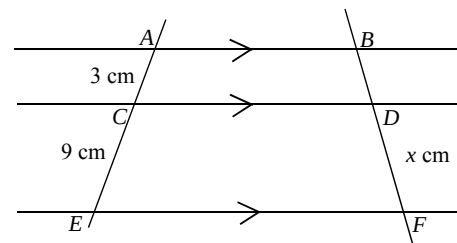
7 The graph of the function $y = f(x)$ is shown in the diagram.



Which of the following gives the exact value for $\int_0^7 f(x) dx$?

- (A) π
- (B) 3
- (C) 5
- (D) $\pi+5$

8 In the diagram $AB \parallel CD \parallel EF$. $AC = 3$ cm, $CE = 9$ cm, $BF = 20$ cm and $DF = x$ cm.

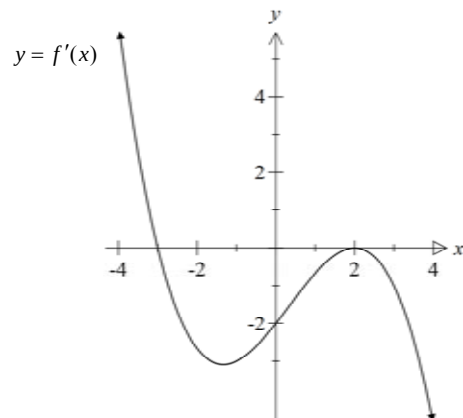


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What is the value of x ?

- (A) 5
- (B) $6\frac{2}{3}$
- (C) 10
- (D) 15

- 9 Consider the graph of the gradient function $y = f'(x)$ below.



Which one of the following statements is true for $y = f(x)$?

- (A) There is a maximum turning point at $x = 2$.
 (B) There is a minimum turning point at $x = -3$.
 (C) There is a horizontal point of inflexion at $x = 2$.
 (D) There is a point of inflexion at $x = 0$.
- 10 Let $c = e^x$. Which expression is equal to $\log_e(c^3)$?

- (A) e^{3x}
 (B) $3x$
 (C) e^{x^3}
 (D) x^3

End of Section I

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet

- (a) Simplify $3x - 2(x - 1)$. 1
- (b) Evaluate $\log_3 5$ correct to three decimal places. 2
- (c) Evaluate $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}$. 2
- (d) Find the primitive of $e^{5x} + 1$. 2
- (e) Given $f(x) = x^2 - 3x - 30$, find the value(s) of x if $f(x) = 10$. 2
- (f) Find the domain and range of $y = \frac{2}{x-3} + 1$. 2
- (g) Find the equation of the parabola with focus $(-3, 1)$ and directrix $y = -1$. 2
- (h) Solve $\frac{2x-1}{4} + 1 = \frac{x}{3}$. 2

Question 12 (15 marks) Use a SEPARATE writing booklet

(a) Solve $\sqrt{3} + \tan x = 0$ for $0^\circ \leq x \leq 360^\circ$. 2

(b) Differentiate the following with respect to x .

(i) $\log_e(3x^2 - 1)$. 2

(ii) $x^2 e^x$. 2

(c) Find $\int x e^{5x^2} dx$. 2

(d) Evaluate $\int_1^3 \frac{2x^3 + x}{x^2} dx$. 3

(e) Find the equation of the normal to the curve $y = 3 \log_e x$ at the point $(e, 3)$. 3

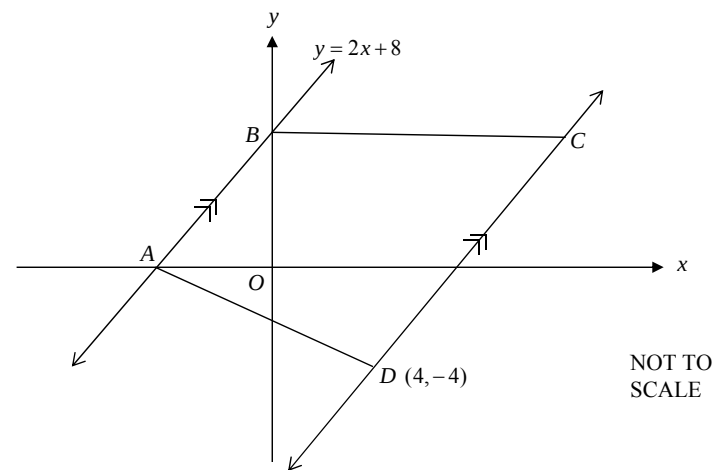
(f) Simplify $\sqrt{\sec^2 A - 1}$. 1

Question 13 (15 marks) Use a SEPARATE writing booklet

(a) In the diagram $ABCD$ forms a trapezium.

Line AB has equation $y = 2x + 8$ and is parallel to line CD .

Line BC is parallel to the x -axis.



(i) Show that the equation of line CD is $2x - y - 12 = 0$. 2

(ii) Show that the perpendicular distance from B to the line CD is $4\sqrt{5}$ units. 2

(iii) Find the co-ordinates of C . 1

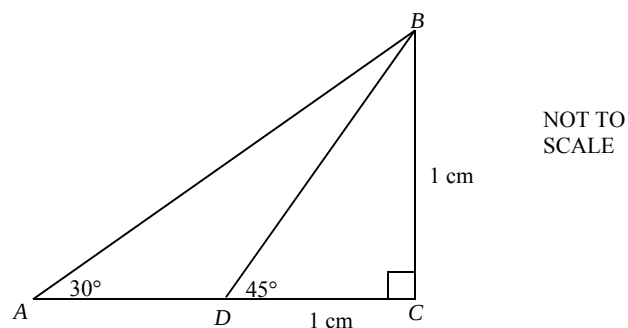
(iv) If $AB = 4\sqrt{5}$ units, find the area of the trapezium $ABCD$. 3

(b) Find the values of k for which $x^2 - 2kx + 1 = 0$ has real and different roots. 2

Question 13 continues on the following page

Question 13 (continued)

- (c) Consider the triangles ABC , BAD and BDC below.
 $\angle ACB = 90^\circ$, $\angle BDC = 45^\circ$, $\angle BAD = 30^\circ$ and $BC = DC = 1$ cm.

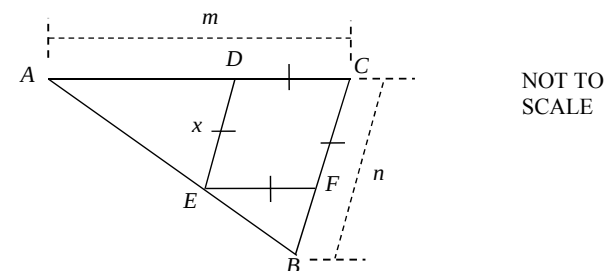


- (i) Using triangle ABC show that the length of AD is $(\sqrt{3} - 1)$ cm. 2
- (ii) Find the length of BD . 1
- (iii) Hence, or otherwise, show that $\sin 15^\circ = \frac{\sqrt{3} - 1}{2\sqrt{2}}$. 2

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet

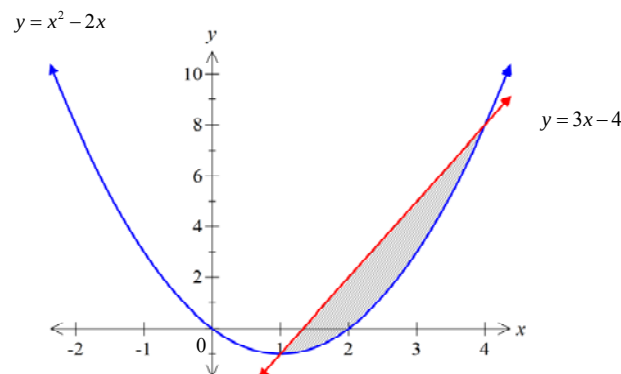
- (a) Consider the series $4 + 7 + 10 + \dots + 301$.
- (i) How many terms are in the series? 2
- (ii) Evaluate the sum of the series. 2
- (b) Use Simpson's rule with 3 function values to find an approximation for $\int_4^8 \frac{1}{\log_e x} dx$. Give your answer correct to 3 significant figures. 2
- (c) Consider the function $y = 2x^3$.
- (i) Draw a neat sketch of this function. 1
- (ii) Find the area between the curve $y = 2x^3$, the x -axis and the lines $x = -1$ and $x = 3$. 3
- (d) The diagram shows a triangle ABC with sides $AC = m$ and $BC = n$. The points D , E and F lie on the sides AC , AB and BC respectively so that $CDEF$ is a rhombus with sides of length x .



- (i) Prove that $\triangle ADE$ is similar to $\triangle EFB$. 2
- (ii) Find an expression for x in terms of m and n . 3

Question 15 (15 marks) Use a SEPARATE writing booklet

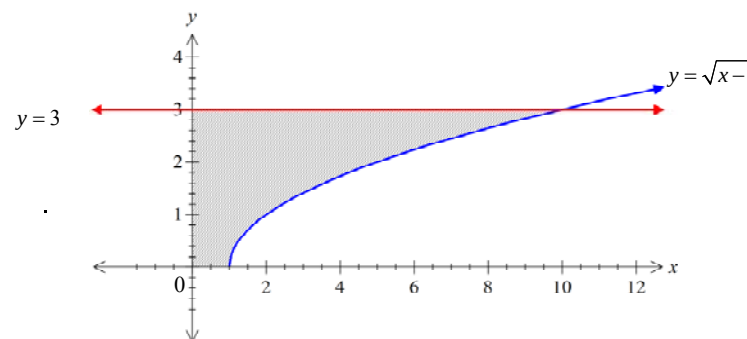
- (a) Evaluate $\int_0^3 \frac{x}{x^2+3} dx$. Express your answer in simplest form. 3
- (b) Consider the curve given by $y = 1 + 3x - x^3$.
- Find the coordinates of the stationary point(s) and determine their nature. 2
 - Find any points of inflexion. 2
 - Hence, sketch the curve showing features found in parts (i) and (ii). 2
(Do not find the x intercepts).
- (c) The shaded region in the diagram is bounded by the curve $y = x^2 - 2x$ and the line $y = 3x - 4$.



- State the points of intersection of the line and the parabola. 1
 - Find the area of the shaded region. 2
- (d) Consider the series $3 + 6p + 12p^2 + 24p^3 + \dots$
- For what values of p does this series have a limiting sum? 1
 - If the limiting sum of this series is $4\frac{1}{2}$, find the value of p . 2

Question 16 (15 marks) Use a SEPARATE writing booklet

- (a) The graph of $y = \sqrt{x-1}$ is shown below. 4



The shaded region in the diagram is bounded by the curve $y = \sqrt{x-1}$, the y -axis and the lines $y = 0$ and $y = 3$.

Find the volume of the solid of revolution formed when the shaded region is rotated about the y axis.

- (b) Mr Smith borrowed \$180 000 to buy a unit. The interest rate was 18% per annum, compounded monthly. He agreed to repay the loan in 20 years with equal monthly repayments. Let the monthly repayments be $\$M$. The amount owing, $\$A_n$, on the loan after the n th monthly repayment is given by the formula

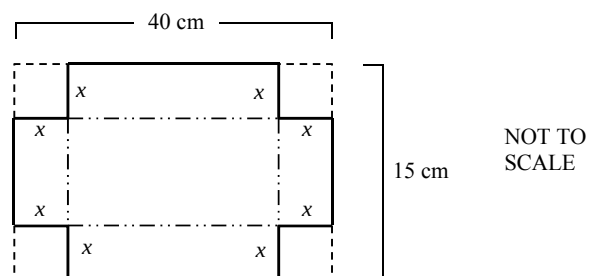
$$A_n = 180\,000(1.015)^n - M(1 + 1.015 + 1.015^2 + \dots + 1.015^{n-1}) \quad (\text{Do not prove this})$$

- If the loan is repaid in full in 20 years show that $\$M$, the monthly repayment is \$2777.96. 2
- Mr Smith decides to round this monthly repayment to the nearest \$1000. How long will it take him to repay the loan, to the nearest month? 3

Question 16 continues on the following page

Question 16 (continued)

- (c) An open rectangular box is to be formed by cutting squares of side length x cm from each corner of a rectangular sheet of metal that has length 40 cm and width 15 cm and folding up the sides.



- (i) Find expressions for the length and breadth of the box in terms of x . 1
- (ii) Show that the volume of the box is given by $V = 600x - 110x^2 + 4x^3$. 2
- (iii) Find the value of x that gives the box its greatest volume. 3

END OF PAPER

YEAR 12 TERM II EXAMINATION

MAY 2016.

Section 1.

1. $0.0157... \div 0.016$ (D)

2. $4m^{-2} \div \frac{1}{2}m^{-1} = 8m^{-2-(-1)}$
 $= 8m^{-1}$
 $= \frac{8}{m}$ (B)

3. $-1 \leq 2x-3 \leq 1$
 $2 \leq 2x \leq 4$
 $1 \leq x \leq 2$ (C)

or Critical values

$2x-3 = 1$ or $2x-3 = -1$
 $2x = 4$ or $2x = 2$
 $x = 2$ or $x = 1$



Test $x = 0$ $|2(0)-3| \leq 1$
 $3 \leq 1$ False

$\therefore 1 \leq x \leq 2$

4. $\frac{3}{2+\sqrt{5}} = a+b$

$\frac{3}{2+\sqrt{5}} \cdot \frac{2-\sqrt{5}}{2-\sqrt{5}} = \frac{6-3\sqrt{5}}{2^2-(\sqrt{5})^2}$
 $= \frac{6-3\sqrt{5}}{4-5}$
 $= \frac{6-3\sqrt{5}}{-1}$

$= -6+3\sqrt{5}$

$a+\sqrt{b} = -6+\sqrt{45}$

$\therefore a = -6$ $b = 45$ (C)

5. $\log_a 24 = \log_a (3 \times 8)$
 $= \log_a 3 + \log_a 8$
 $= \log_a 3 + \log_a 2^3$
 $= \log_a 3 + 3\log_a 2$
 $= q + 3p$ (A)

6. $2x^2 - 4x + 7 = 0$

$x + \beta = -\frac{b}{a}$
 $= \frac{4}{2}$
 $= 2$

$x\beta = \frac{c}{a}$
 $= \frac{7}{2}$

$x^2 + \beta^2 = (x+\beta)^2 - 2x\beta$
 $= 2^2 - 2 \times \frac{7}{2}$
 $= 4 - 7$
 $= -3$ (A)

7. $\int_0^1 f(x) dx = A_{\text{square}} + A_{\text{area } \Delta}$
 $= 4 + -\frac{1}{2} \times 1 \times 2$
 $= 4 - 1$
 $= 3$ (B)

8. $\frac{x}{20} = \frac{9}{12}$

$x = \frac{9}{12} \times 20$
 $= 15$ (D)

-2-

9. C

10. $\log_e (e^3) = \log_e (e^x)^3$
 $= \log_e e^{3x}$
 $= 3x \log_e e$
 $= 3x$ (B)

1 D	6 A
2 B	7 B
3 C	8 D
4 C	9 C
5 A	10 B

Question 11.

(a) $3x - 2(x-1)$
 $= 3x - 2x + 2$
 $= x + 2$ (1)

(b) $\log_3 5 = \frac{\log_e 5}{\log_e 3}$ (2)
 $= 1.4649...$
 ≈ 1.465 (3dp)

(c) $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}$ (2)
 $= \lim_{x \rightarrow 2} \frac{(x-2)(x^2+2x+4)}{(x-2)}$
 $= 2^2 + 2(2) + 4$
 $= 12$

(d) $\frac{dy}{dx} = e^{5x} + 1$ (2)

$y = \frac{e^{5x}}{5} + x + c$

(e) $f(x) = 10$ (2)
 $x^2 - 3x - 30 = 10$
 $x^2 - 3x - 40 = 0$
 $(x-8)(x+5) = 0$
 $x = 8$ or -5

(f) Domain: all real x , $x \neq 3$ (2)
 Range: all real y , $y \neq 1$

(g) $(x+3)^2 = 4y$ $V = (-3, 0)$ (2)

(h) $\frac{2x-1}{4} + 1 = \frac{x}{3}$ (x+2)
 $3(2x-1) + 12 = 4x$ (2)
 $6x - 3 + 12 = 4x$
 $2x = -9$
 $x = -4\frac{1}{2}$

Question 12.

(a) $\sqrt{3} + \tan x = 0$
 $\tan x = -\sqrt{3}$

related $L = 60^\circ$

$x = 180^\circ - 60^\circ$, $360^\circ - 60^\circ$

$x = 120^\circ$ or 300° (2)



(b)(i) $y = \log(3x^2 - 1)$ [2]

$$\frac{dy}{dx} = \frac{1}{3x^2 - 1} \times 6x$$

$$= \frac{6x}{3x^2 - 1}$$

(ii) $y = x^2 e^x$ [2]

$$\begin{cases} u = x^2 & v = e^x \\ u' = 2x & v' = e^x \end{cases}$$

$$y' = vu' + uv'$$

$$= e^x \times 2x + x^2 \times e^x$$

$$= 2xe^x + x^2 e^x$$

$$= xe^x(2 + x)$$

(c) $\int x e^{5x^2} dx = \frac{1}{10} e^{5x^2} + c$ [2]

(d) $\int_1^3 \left(\frac{2x^3}{x^2} + \frac{1}{x} \right) dx$

$$= \int_1^3 \left(2x + \frac{1}{x} \right) dx$$

$$= \left[\frac{2x^2}{2} + \ln x \right]_1^3$$

$$= [x^2 + \ln x]_1^3$$

$$= [(3^2 + \ln 3) - (1^2 + \ln 1)]$$

$$= 9 + \ln 3 - 1$$

$$= \underline{8 + \ln 3}$$

(e) $y = 3 \log_e x$ [3]

$$\frac{dy}{dx} = \frac{3 \times 1}{x}$$

$$= \frac{3}{x}$$

When $x = e$

$$\frac{dy}{dx} = \frac{3}{e} = m_T$$

$$\therefore m_N = -\frac{e}{3}$$

Eqn of normal. $(e, 3)$

$$y - 3 = -\frac{e}{3}(x - e)$$

$$3y - 9 = -ex + e^2$$

$$ex + 3y - 9 - e^2 = 0$$

(f) $\sqrt{\sec^2 A - 1} = \sqrt{\tan^2 A}$ [1]

$$= \tan A$$

$$[1 + \tan^2 A = \sec^2 A]$$

$$\therefore \sec^2 A - 1 = \tan^2 A$$

Question 13.

(i) $m_{AB} = 2 \therefore m_{CD} = 2$ (AB || CD) [2]

Eqn CD.

$$y + 4 = 2(x - 4)$$

$$y + 4 = 2x - 8$$

$$0 = 2x - y - 12$$

(ii) $2x - y - 12 = 0$

$$a = 2 \quad b = -1 \quad c = -12$$

$$B(0, 8)$$

pd = $\frac{|ax + by + c|}{\sqrt{a^2 + b^2}}$ [2]

$$= \frac{|2 \times 0 + -1 \times 8 - 12|}{\sqrt{4 + 1}}$$

$$= \frac{|-20|}{\sqrt{5}}$$

$$= \frac{20}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}}$$

$$= \frac{20\sqrt{5}}{5}$$

$$= \underline{4\sqrt{5} \text{ units}}$$

(iii) $A + C, y = 8$ [1]

Sub into $2x - y - 12 = 0$

$$2x - 8 - 12 = 0$$

$$2x = 20$$

$$x = 10$$

$$\therefore C = (10, 8)$$

(iv) $d_{BC} = \sqrt{(10 - 4)^2 + (8 + 4)^2}$ [3]

$$= \sqrt{36 + 144}$$

$$= \sqrt{180}$$

$$= 3\sqrt{20}$$

$$= 3 \times 2\sqrt{5}$$

$$= \underline{6\sqrt{5}}$$

$$\therefore A = \frac{1}{2} h(a + b)$$

$$\text{trap} = \frac{1}{2} \times 4\sqrt{5} (4\sqrt{5} + 6\sqrt{5})$$

$$= 2\sqrt{5} \times 10\sqrt{5}$$

$$= \underline{100 \text{ units}^2}$$

(b) real & diff roots when $\Delta > 0$

$$\Delta = b^2 - 4ac$$
 [2]

$$= (-2k)^2 - 4(1)$$

$$= 4k^2 - 4$$

$$\therefore 4k^2 - 4 > 0$$

$$k^2 - 1 > 0$$

$$(k - 1)(k + 1) > 0$$

$$k = \pm 1$$

$$\leftarrow \text{---} \overset{x}{\underset{-1 \quad 0 \quad 1}{\text{---}}} \rightarrow$$

test $\therefore = 0 \quad 0 - 1 > 0$

$$-1 > 0 \text{ False}$$

$$\therefore k < -1 \text{ or } k > 1$$

(c) In ΔABC

(i)

$$\tan 30^\circ = \frac{1}{AC}$$

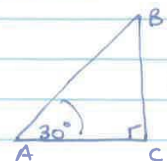
$$AC = \frac{1}{\tan 30^\circ}$$

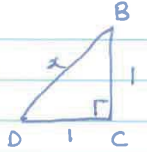
$$= \frac{1}{\frac{1}{\sqrt{3}}}$$

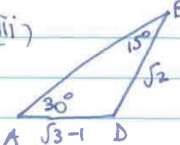
$$= \sqrt{3}$$

$$\therefore AD = AC - DC$$

$$= (\sqrt{3} - 1) \text{ cm}$$



(ii)  $x^2 = 1^2 + 1^2$
 $= 2$
 $x = \sqrt{2}$
 $BD = \sqrt{2}$ [1]

(iii)  sine rule: [2]
 $\frac{\sin 15^\circ}{\sqrt{3}-1} = \frac{\sin 30^\circ}{\sqrt{2}}$
 $\sin 15^\circ = \frac{1}{\sqrt{2}} (\sqrt{3}-1)$
 $\sin 15^\circ = \frac{\sqrt{3}-1}{2\sqrt{2}}$

Question 14.

(i) $a = 4$ $d = 3$ $T_n = 301$

$T_n = a + (n-1)d$
 $301 = 4 + (n-1)3$
 $301 = 4 + 3n - 3$
 $300 = 3n$
 $100 = n$

there are 100 terms

(ii) $S_{100} = ?$ [2]

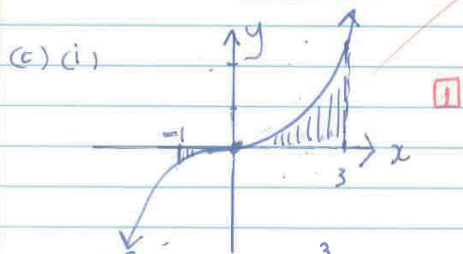
$S_n = \frac{n}{2} [a + l]$
 $S_{100} = \frac{100}{2} [4 + 301]$
 $= 15250$

(b) $y = \frac{1}{\log x}$ [2]

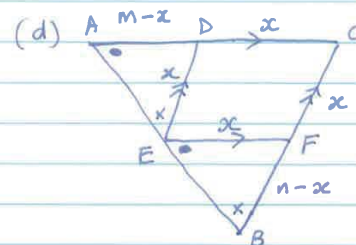
x	4	6	8
y	$\frac{1}{\ln 4}$	$\frac{1}{\ln 6}$	$\frac{1}{\ln 8}$

$\int_a^b f(x) dx \approx \frac{b-a}{6} [f(a) + 4f(\frac{a+b}{2}) + f(b)]$

$\int_4^8 \frac{1}{\log x} dx \approx \frac{8-4}{6} [\frac{1}{\ln 4} + 4 \cdot \frac{1}{\ln 6} + \frac{1}{\ln 8}]$
 $= 2.289...$
 ≈ 2.29 (3 sf)



(ii) $A = \left| \int_{-1}^0 2x^3 dx \right| + \int_0^3 2x^3 dx$
 $= \left| \left[\frac{2x^4}{4} \right]_{-1}^0 \right| + \left[\frac{2x^4}{4} \right]_0^3$ [3]
 $= \left| \left[\frac{x^4}{2} \right]_{-1}^0 \right| + \left[\frac{x^4}{2} \right]_0^3$
 $= \left| \frac{0}{2} - \frac{(-1)^4}{2} \right| + \left[\frac{3^4}{2} - 0 \right]$
 $= \left| -\frac{1}{2} \right| + \frac{81}{2}$
 $= \frac{82}{2}$
 $= 41 \text{ units}^2$



In $\triangle ADE$ and $\triangle EFB$
 $\angle DAE = \angle FEB$ (corresponding $\angle$ s,
 $AC \parallel EF$)
 $\angle DEA = \angle FBE$ (corresponding $\angle$ s,
 $ED \parallel BC$)
 $\therefore \triangle ADE \sim \triangle EFB$ (equiangular)

(ii) $\therefore \frac{x}{m-x} = \frac{n-x}{x}$ (matching sides of similar Δ s in proportion)
 $x^2 = (m-x)(n-x)$
 $x^2 = mn - mx - nx + x^2$
 $0 = mn - mx - nx$
 $= mn - x(m+n)$
 $x(m+n) = mn$
 $x = \frac{mn}{m+n}$ [3]

Question 15. [3]

(a) $\int_0^3 \frac{x}{x^2+3} dx = \left[\frac{1}{2} \ln(x^2+3) \right]_0^3$
 $= \frac{1}{2} \ln(3^2+3) - \frac{1}{2} \ln(0^2+3)$
 $= \frac{1}{2} \ln 12 - \frac{1}{2} \ln 3$
 $= \frac{1}{2} \left[\ln \frac{12}{3} \right]$
 $= \frac{1}{2} \ln 4$
 $= \frac{1}{2} \times 2 \ln 2$
 $= \ln 2$

(b) $y = 1 + 3x - x^3$
 (i) $\frac{dy}{dx} = 3 - 3x^2$ [2]
 $\frac{d^2y}{dx^2} = -6x$

Stat. pts when $\frac{dy}{dx} = 0$

$3 - 3x^2 = 0$
 $3 = 3x^2$
 $x^2 = 1$
 $x = \pm 1$

stat pts $(-1, -1)$ and $(1, 3)$

$x = -1$ $y = 1 + (3(-1)) - (-1)^3$
 $= 1 - 3 + 1$
 $= -1$
 $x = 1$ $y = 1 + 3 - 1$
 $= 3$

When $x = -1$ $\frac{d^2y}{dx^2} = -6(-1)$
 $= 6 > 0$

$\therefore$ Min. turning pt at $(-1, -1)$

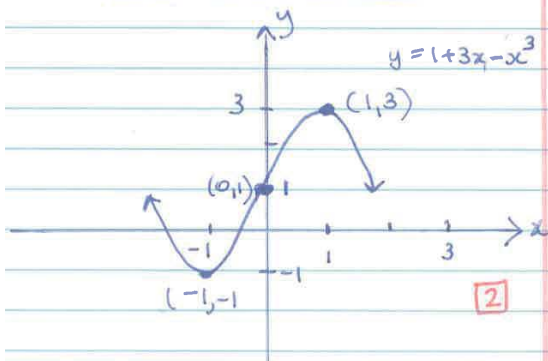
when $x = 1$ $\frac{d^2y}{dx^2} = -6(1)$
 $= -6 < 0$

$\therefore$ Max. turning pt $(1, 3)$

(ii) pts of inflection when $\frac{d^2y}{dx^2} = 0$
and change of concavity
 $\frac{d^2y}{dx^2} = -6x$
 $-6x = 0$
 $\therefore x = 0$

x	$-\frac{1}{2}$	0	$\frac{1}{2}$
$\frac{d^2y}{dx^2}$	3	0	-3

$\therefore$ pt of inflexion at $(0, 1)$



[2]

[2]

(c) (i) Pts of intersection
 $(1, -1)$ and $(4, 8)$ [1]

$$\begin{aligned} \text{(ii)} A &= \int_1^4 [f(x) - g(x)] dx \\ &= \int_1^4 [(3x-4) - (x^2-2x)] dx \\ &= \int_1^4 (3x-4-x^2+2x) dx \\ &= \int_1^4 (5x-4-x^2) dx \quad [2] \\ &= \left[\frac{5x^2}{2} - 4x - \frac{x^3}{3} \right]_1^4 \\ &= \left[\left(\frac{5(4)^2}{2} - 4(4) - \frac{4^3}{3} \right) - \left(\frac{5}{2} - 4 - \frac{1}{3} \right) \right] \\ &= 4.5 \text{ units}^2 \end{aligned}$$

(d) $3 + 6p + 12p^2 + \dots$
 $r = 2p$

For a limiting sum

$$-1 \leq r \leq 1$$

$$-1 \leq 2p \leq 1$$

$$-\frac{1}{2} \leq p \leq \frac{1}{2}$$

[1]

(ii) $S_{\infty} = \frac{a}{1-r}$

$$4\frac{1}{2} = \frac{3}{1-2p}$$

$$1-2p = \frac{3}{4\frac{1}{2}}$$

[2]

$$-2p = \frac{3}{4\frac{1}{2}} - 1$$

$$p = \frac{1}{6}$$

Question 16.

$$V = \pi \int x^2 dy$$

$$= \pi \int_0^3 (y^2+1)^2 dy$$

$$= \pi \int_0^3 (y^4 + 2y^2 + 1) dy$$

$$= \pi \left[\frac{y^5}{5} + \frac{2y^3}{3} + y \right]_0^3$$

$$= \pi \left[\left(\frac{3^5}{5} + \frac{2(3)^3}{3} + 3 \right) - \left(\frac{0}{5} + 2(0) + 0 \right) \right]$$

$$= \frac{348\pi}{5} \text{ units}^3$$

$$\begin{aligned} y &= \sqrt{x-1} \\ y^2 &= x-1 \\ y^2+1 &= x \\ \therefore x^2 &= (y^2+1)^2 \\ &= y^4 + 2y^2 + 1 \end{aligned}$$

[4]

(b) $A_{240} = 180\,000 (1.015)^{240} - M (1 + 1.015 + \dots + 1.015^{239})$
 $\nearrow S_n, n=240, a=1, r=1.015$

$$S_n = \frac{a(r^n-1)}{r-1}$$

[2]

$$S_{240} = \frac{1.015^{240} - 1}{0.015}$$

$A_{240} = 0$ after 24yrs. $\therefore 0 = 180\,000 (1.015)^{240} - M \frac{(1.015^{240} - 1)}{0.015}$
 $M \frac{(1.015^{240} - 1)}{0.015} = 180\,000 (1.015)^{240}$

$$M = 180\,000 (1.015)^{240} \div \frac{1.015^{240} - 1}{0.015}$$

$$= \$2\,717.96$$

$$M = \$3000$$

$$(ii) \text{ From (i) } A_n = 180\,000 (1.015)^n - 3000 \left(\frac{1.015^n - 1}{0.015} \right)$$

$$A_n = 0 \quad 0 = 180\,000 (1.015)^n - 200\,000 (1.015^n - 1)$$

$$0 = 180\,000 (1.015)^n - 200\,000 (1.015)^n + 200\,000$$

$$0 = -20\,000 (1.015)^n + 200\,000$$

$$20\,000 (1.015)^n = 200\,000 \quad (\div 20\,000)$$

$$1.015^n = 10$$

$$\log_e (1.015)^n = \log_e 10$$

$$n \log_e (1.015) = \log_e 10$$

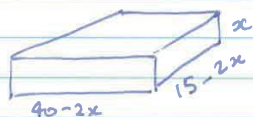
$$n = \frac{\log_e 10}{\log_e (1.015)}$$

$$= 154.65$$

$$= \underline{155 \text{ mths (nearest mth)}}$$

[3]

(c)



$$(i) \text{ length} = 40 - 2x$$

$$\text{breadth} = 15 - 2x$$

[1]

$$(ii) \quad V = lwh$$

$$= x(40 - 2x)(15 - 2x)$$

$$= x[600 - 80x - 30x + 4x^2]$$

$$= \underline{600x - 110x^2 + 4x^3}$$

[2]

$$(iii) \quad \frac{dV}{dx} = 600 - 220x + 12x^2$$

$$\frac{d^2V}{dx^2} = -220 + 24x$$

[3]

$$\text{Stat pts when } \frac{dV}{dx} = 0$$

$$600 - 220x + 12x^2 = 0$$

($\div 4$)

$$3x^2 - 55x + 150 = 0$$

$$x = \frac{55 \pm \sqrt{55^2 - 4(3)(150)}}{6}$$

$$= \frac{55 \pm 35}{6}$$

$$x = \frac{90}{6} \text{ or } x = \frac{20}{6}$$

$$\underline{x = 15 \text{ or } x = 3\frac{1}{3}}$$

$$\text{When } x = 15 \quad \frac{d^2V}{dx^2} = -220 + 24 \times 15$$

$$= 160 > 0$$

$\therefore$ Min

$$\text{When } x = 3\frac{1}{3} \quad \frac{d^2V}{dx^2} = -220 + 24 \times 3\frac{1}{3}$$

$$= -120 < 0 \quad \therefore \text{Max}$$

$\therefore$ Max volume when $x = 3\frac{1}{3} \text{ cm.}$